



INFORM

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DESTINATION FUTURE.



Title page.

Based in the Italian town of Schio, SAF Bianco manufactures high-quality plastic packing for renowned brands in its homemarket.

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Dear Readers,

Since 15 January 2015, the situation for manufacturing SMEs like ours has worsened dramatically in Switzerland. In order to continue to survive on the international stage, we are forced to do more than our competitors from Europe, Asia, China and the USA.

There is no other option. Our moulds are used almost exclusively in companies outside Switzerland. Of course, their quality is praised and their reliability appreciated. But in future this will no longer be enough to secure the growth required for the stability of the company.

There are many different underlying causes. From the economic perspective, we are obliged to safeguard or expand our global market share in the long term. We must even out economic, seasonal and structural fluctuations in demand and learn how to deal with obstacles to trade. From the marketing angle, we won't be able to avoid increasing our presence on the market and vis-à-vis our customers. Furthermore, we would be well advised to optimise our understanding of the mechanisms in the different sales regions and build on our knowledge of socio-cultural and political-legal framework conditions.

Our declared goal is to continue to maintain our overall portfolio of top-notch moulds and exemplary consulting activities and services on the highest level. This is what prompted us to consider setting up a sales organisation outside Switzerland. As a mid-sized company we cannot do this completely on our own and that is why we are currently taking a long, hard look at cooperation with an international partner. We will keep you informed.

Your Otto Hofstetter

HARD TO BEAT: A GLOBAL PLAYER.



ALPLA Werke Alwin Lehner GmbH & Co KG is a world-renowned company. The family-run company ranks amongst the **leading companies for plastic packaging**. On the production side, it places its trusts, amongst other things, in the moulds of Otto Hofstetter AG. *inform* paid a visit to its headquarters in Hard and talked to **CEO Günther Lehner** and **Dietmar Marin**, Business Unit Manager Injection & Stretch Blow Moulding & Recycling.



Under the leadership of Dietmar Marin, the PET sector has grown markedly in recent years.

ALPLA was established in 1955. How did that happen?

Günther Lehner: My father and my uncle started from scratch without any prior knowledge of plastics. They were pioneers. Today, I think we would call this a start-up. In the 1950s, plastics gained a real foothold in the packaging industry and were increasingly in demand. With their expertise, they advanced the development of the technology also in the areas of producing moulds and machinery. They not only laid the foundation stone for our company but, at the same time, they also launched the momentum that led to our position today as a technology leader.

**"OUR CUSTOMERS EXPECT
THE SAME TREATMENT IN EACH
COUNTRY WHEN THEY COOPERATE
WITH ALPLA."**

**Dietmar Marin, Business Unit Manager Injection
& Stretch Blow Moulding & Recycling.**

Today, ALPLA is one of the leading global companies in the plastic packaging industry. What is your recipe for success?

Günther Lehner: Since its foundation, ALPLA has geared its activities to the requirements of its customers. In parallel to the developments in plastics, bottles and caps, the company has grown alongside its customers. The company was quick

to grasp that it could only be a valuable partner for its customers if it understands their business. This philosophy still applies today. Of course, we did not shy away from entrepreneurial risks over the last 60 years. We continuously invested in further developments and carried the company forward with strong partners and a clear focus. Our goal is not to engage in mass production. We are far more interested in carrying topics over into practice and developing solutions which are completely new in this form on the market.

What was the trigger for internationalisation?

Günther Lehner: Bottles are empty containers which means that you transport air "around the world". In our opinion, this did not seem to make economic sense in the long term. This very quickly prompted us to extend our activities to other regions. Today, we are a global group with 159 plants in 42 countries. We have always endeavoured to consistently implement what we started.

What makes a company competitive today?

Dietmar Marin: The benefit for the customer is the driving force behind all our projects. We combine this with efficient, productive processes, we develop marketing concepts and establish, as far as possible, cost-optimised production. Thanks to our technical expertise in plastic bottles, caps and

"FROM THE VERY OUTSET,
ALPLA HAS GEARED ITS ACTIVITIES
TO THE REQUIREMENTS OF
ITS CUSTOMERS."

Günther Lehner, CEO.

Günther Lehner is part of the second generation of the family which has run the company since it was established in 1956.

preforms, we can offer our customers an all-inclusive package from the idea right through to the finished supply chain solution. The ability to compete is down to the combination of these factors. This is what enables us to keep one step ahead.

Günther Lehner: Besides comprehensive consulting and customised solutions for customers, price plays a decisive role in the end. Particularly in the preform business where you are dealing with pure "commodities". In this environment, we have to score points with good service and impeccable quality to secure out customers' loyalty. This proximity gives us an opportunity to finely tune the solutions for our customers.

How do ideas take shape?

Dietmar Marin: There are many different ways. Sometimes, we receive enquiries about an existing product. In other cases, the development of a product is part of the overall project. However, very frequently a customer's marketing department does its homework and obtains a very clear idea of what it would like to implement from market research. For marketing experts, we are the partner who can support them in the successful execution of their plan. Projects like these are attractive for us because we can be part of the creative process. Although I have to say that optimising existing products is a fascinating challenge every time. Parallel to this, we also do corresponding market research and

develop new, innovative packaging solutions which we then offer to our customers as the alternative to optimising existing packaging.

"CERTIFICATES ARE NOT A DECISIVE
CRITERION FOR US. WHAT IS
REALLY IMPORTANT IS ACTUAL
PERFORMANCE."

Dietmar Marin, Business Unit Manager Injection
& Stretch Blow Moulding & Recycling.

ALPLA is characterised by strong entrepreneurial values. How do you ensure this spirit prevails at all your sites?

Günther Lehner: We have a clear mission and we strive to get all our employees on board throughout the entire process. We cannot get round taking cultural differences into account. We western Europeans – or to be more precise Alemannians – cannot simply assume that everyone thinks like we do. In the past, paying considerable attention to our staff members has worked very well for us. We have always endeavoured to highlight what is important to ensure we can continue doing business.

Dietmar Marin: Of course, our customers in each country expect the same treatment when they cooperate with ALPLA. This means that, for example, we cannot make any conces-



sions when it comes to quality just because their culture differs slightly from our values. We cooperate a great deal with international customers and there is no leeway. The balance between understanding differences and clear framework conditions constitutes a major challenge. In my opinion, we have mastered this very well up to now.

ISO, FSSC, HACCP: you impose very high quality demands on yourself. What qualities are required of a company that wishes to become a partner of ALPLA?

Dietmar Marin: We have basic requirements our partners have to meet. One important criterion is a high degree of reliability. Furthermore, we set our quality standards on the highest level to ensure that the processes can be repeated without even the tiniest deviation. Certificates are not a decisive criterion for us. What is really important is actual performance. Furthermore, we are interested in partners that actively input their knowledge and show themselves to be absolutely reliable.

Do you source injection moulds from Otto Hofstetter AG? Since when have you counted on this manufacturing company in Uznach?

Günther Lehner: For some years now. Back then, ALPLA was the largest manufacturer of PVC packaging in Europe. From one year to the next, the big Swiss retailers decided to no longer use PVC in their packaging. We had to press

ALPLA started with a modest workshop and grew, in the course of its 61-year history, into one of the leading companies for plastic packaging with plants in 42 countries around the globe.



ahead with the conversion to PET as speedily as possible and we were given vigorous support by Otto Hofstetter AG. As both companies were pioneers in the field of technology, this project worked out very well and in the second plant of ALPLA there was already a mould from Uznach.

**"AS A FAMILY-RUN COMPANY,
YOU HAVE THE LUXURY OF BEING ABLE
TO THINK IN THE LONG TERM AND
I BELIEVE THIS SECURES US A MAJOR
ADVANTAGE."**

Günther Lehner, CEO.

How satisfied are you with Otto Hofstetter AG?

Dietmar Marin: As a mould supplier, Otto Hofstetter AG is a good partner and we appreciate this cooperation. Both quality and reliability are exactly as they should be. We would very much welcome greater flexibility when it comes to machine platforms. In addition, we would be interested in jointly cooperating in less technology-driven areas.

Like Otto Hofstetter AG, ALPLA is a family-run company. In terms of global development: what prospects do family-run companies have?

Günther Lehner: We are and wish to remain a wholly-owned family company. Of course, the pressure is not getting any smaller and the requirements are not becoming any less



ALPLA-Kids is just one of many diverse commitments by ALPLA to its employees and regional developments.

exacting. As a family-run company, you have the luxury of being able to think in the long term and I believe this secures us a major advantage. The crisis years 2008 and 2009 have shown that customers appreciate the way in which companies like ours act in difficult times. The future course has been laid out and the third generation is already in the starting blocks.

Ethical values are of central importance for you. In your opinion, what are the biggest challenges that will have to be mastered jointly in industry?

Günther Lehner: The way in which we deal with each other is important. Right now, an example from Germany illustrates what can happen when one of the parties comes up short in this area. All the sides are called on to create the framework conditions for reasonable cooperation. Moral action is centre stage of ethics. Even if it constitutes a major challenge, we will stick to this vis-à-vis our employees, partners and customers.

Mr Lehner, Mr Marin, thank you very much for this very informative and frank interview. I wish you every continued success.



ALPLA

ALPLA Werke Alwin Lehner GmbH & Co KG

Mockenstrasse 34, 6971 Hard, Austria

Products: Plastic packaging (packaging systems, bottles, caps and injection-moulded components, preforms, tubes)

Year of establishment: 1955

Headcount: 16,500

Production sites: 159 in 42 countries

Sales revenues 2015: 3.3 billion euro

COOL DOWN.

OH-XCool® achieves tangible efficiency enhancement in the production of PET preforms. The in-house developments of Otto Hofstetter AG significantly improve the cooling of threads and support rings.

15x

THE TIME SAVED ANNUALLY THANKS TO OH-XCOOL® EQUATES TO 15 TRIPS AROUND THE EARTH (AT A SPEED OF 800 KM/H AND A HEIGHT OF 10,000 METRES ABOVE SEA LEVEL)

300

MILLIONS PREFORMS
ANNUALLY

96

-CAVITY OHAG MOULD

-0,9

SECOND REDUCTION IN
THE CYCLE TIME WITH
OH-XCOOL®
(PCO 28 SHORT)

Sample calculation: Annual production of 300 millions preforms with a 96-cavity OHAG mould. When you multiple the reduction in cycle time of 0.9 seconds for the production of PCO 28 short preforms thanks to OH-XCool® by the annual production volume, the time saved is enough for a plane to fly around the earth 15 times.

It is common knowledge that in an injection mould, the first attempts strive to inject the material very quickly, without wasting any of the material, and to avoid any loss of temperature in the molten mass as much as possible. Once this process has been completed, the next immediate, follow-on step in high-performance moulds like the ones needed, for instance, for the production of PET preforms is cooling of the injection-moulded part. The ever shorter cycle times are, therefore, dependent on a very even temperature in the mould and absolutely precision-adjusted process parameters in the control of the injection moulding machine. When a particularly high volume of plastic has to be cooled in the thread and support ring of the preform, and this is then the determining factor for cycle time, attention mainly focuses on this aspect.

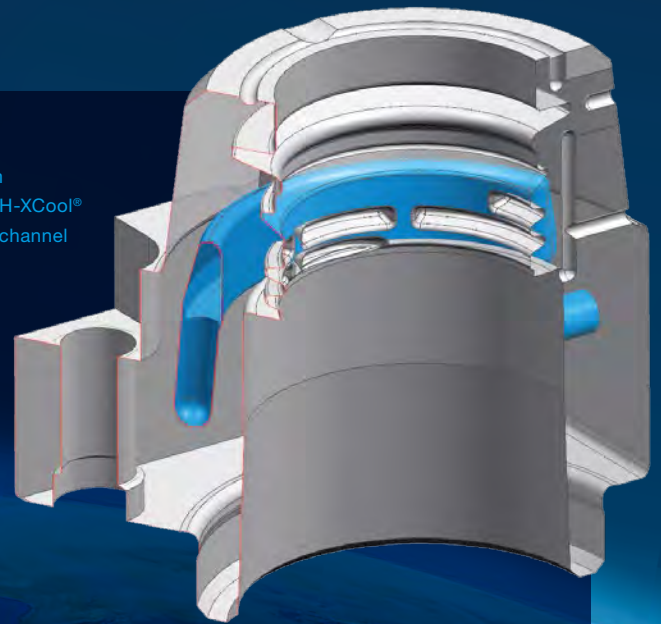
Cooling in line with conventional standards.

What goes for many other types of moulds goes for the PET moulds manufactured by Otto Hofstetter AG, too. The mould components are fitted, if possible, with efficient cooling. The generally valid maxim in the construction of these steel parts is that the cooling channel should be positioned as close as possible to the mould contour of the injection-moulded part. They are manufactured using standard operating procedures like, for instance, drilling or milling. In most cases, this is sufficient to achieve the required cooling capacity.

The thread takes things up a notch.

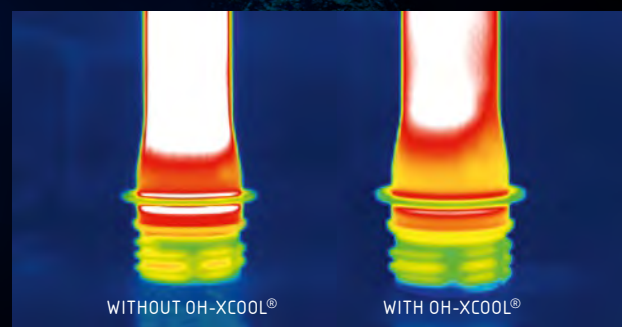
However, cooling in the thread insert of a PET mould imposes considerably stiffer and highly specific requirements for the production of cooling channels. The dimensions of these components are significantly smaller and the amount of steel to be processed as the basic material is available in a far more limited quantity. Consequently, only very small bore holes can be drilled. This leads to a limited cooling capacity and, in addition, to a far higher risk of contamination. Furthermore, optimum convergence with the mould contour is only partially possible.

Diagram
of the OH-XCool®
cooling channel



OH-XCool® adds a new twist.

Basically, this is where the OH-XCool® technology developed by technicians in Uznach steps in. Thanks to a special manufacturing process, they have succeeded in integrating a wrap-around cooling channel close to the contour in an optimum position in terms of the thread and support ring. What is more, this exclusive method offers yet another advantage. Thanks to the position of the bore hole it is also possible to lay a wider channel, which is the precondition for a far higher cooling capacity. This leads to a reduction in cycle time which more than pays for itself. Depending on the type of preform, the process is shortened by up to 15 per cent. OH-XCool® is available as an option for all new generation moulds of Otto Hofstetter AG.



In the thermal images, a clear reduction in the "hot points" in the thread is visible.

CYCLETIME

PREFORM / THREAD TYPE	WEIGHT	WALL THICKNESS	STANDARD	OH-XCOOL®
Bericap 30-37	34.0 g	3.57 mm	14.6"	13.3"
Twist off	29.8 g	2.70 mm	10.5"	10.0"
PCO 28 short	18.15 g	2.47 mm	9.2"	8.3"
PCO 28 Alcoa	41.25 g	3.71 mm	14.2"	12.5"

A portrait of Massimiliano Bianco, a middle-aged man with dark, slightly messy hair and a light beard. He is smiling and looking towards the camera. He is wearing a dark navy blue suit jacket over a white dress shirt and a dark tie with a small, repeating geometric pattern. His hands are clasped together on a dark, reflective surface in front of him. He is wearing a silver-toned wristwatch with a dark face on his left wrist. The background is a bright, out-of-focus office environment.

SLOW
BUT
STEADY.

For Massimiliano Bianco, working in accordance with the principle "form follows function" is vital when creating new products.

SAF Bianco manufactures high-quality plastic packaging for renowned food producers in Italy. The family-owned company, which is based in the northern Italian town of Schio, has chosen Otto Hofstetter AG as the firm's partner for its injection moulding tools. Massimiliano Bianco, who has officially held the role of CEO since 1 September 2016, explains in an interview how this choice will prove beneficial, and provides an insight into the characteristics of the Italian packaging industry.

Mr Bianco, what does "SAF" stand for in your company name?

The abbreviation stands for "stampaggio articoli fenolici", which refers to the pressing of products with phenoplast, a thermoplastic material developed in the 1950s.

SAF Bianco celebrated its 70th anniversary at the beginning of the year. How did your company's success story begin?

My grandfather spent a short time in Brussels during the 1940s. It was there that he got to know the injection moulding business and acquired the associated skills. In 1945, following his return to Italy, he started production at home in his own small workshop. I am thus a member of the third generation. The company is currently managed together by two generations: the team is made up of the fathers Diego and Eddo and their sons Massimiliano, Simone and Raffaele.

What does SAF look like today?

SAF employs 37 staff members, and our fleet of machinery includes 14 injection moulding machines, three sleeve systems and two offset printing machines. Our annual production volume stands at around 350 million packaging units which take various forms.

Which markets do you serve?

Italy is our core market. We specialise in the production of packaging for sauces, desserts, milk and cheese products and snacks. Many renowned brands in the domestic food sector, including Parmalat, Galbani, Sterilgarda, Emmi and Giovanni Rana, count on our products.



Those are major brands. What is SAF Bianco's position on the Italian market?

I believe it is fair to say that we are one of the leading companies in our sector, in particular in terms of the quality of our equipment. Our leading position when it comes to quality can also certainly be attributed to the fact that we purchase our injection moulding tools abroad and can, among other factors, rely on the expertise of Otto Hofstetter AG.

**"I BELIEVE IT IS FAIR TO SAY
THAT WE ARE ONE OF THE LEADING
COMPANIES IN OUR SECTOR."**

Massimiliano Bianco, CEO.

In what areas does your company specialise?

Thanks to our modern and efficient fleet of machinery, the precision of our tools and the quality materials we use, we are able to produce very thin-walled packaging that is nevertheless dimensionally stable. Light packaging means that less raw material is required, which in turn has a positive impact on the production price and improves our environmental performance. The aforementioned relates to the production side of our operations. What also sets us apart, however, is our design department with our designer Luca Floreanini. This allows us to support our customers from the initial moment an idea is formed to the production of the finished packaging.

What form does this process take?

The customer contacts us and we arrange a briefing. We want to find out first-hand what the customer's wishes and intentions are in connection with the product. I then sit down with our designer and we draw up an initial design. Working in accordance with the principle of "form follows function", first a technical drawing is prepared, and it is only then, on this basis, that the design is developed. To enable customers to handle and assess their product, we produce a prototype on a 3-D printer. Should

we get the green light, we then make a pilot mould and produce production samples. This process provides the customer with maximum security and generally only takes six months.

You have listed a number of major brands you serve. Why do these companies give preference to SAF over your competitors?

One aspect is, without a doubt, the benefits we offer from a technical and creative perspective. I find, however, that our customers also value our philosophy. At SAF Bianco, we do not talk about "customers" or "suppliers". We work together with partners, in both directions. I will give you an example: Rana needs sauce cups and gives us a call. We then contact Otto Hofstetter AG, which builds the tool for the Rana packaging. It works as a circle, and we are more effective if we operate within this circle as a partner. After all, reliability, speed and flexibility are extremely important nowadays. And we can guarantee these qualities with our philosophy.

What are the biggest challenges when working as a partner to food producers?

Food safety is the key issue and demands a great deal of attention. On the one hand, international standards have to be meticulously adhered to and processes need to be recertified on an ongoing basis. On the other hand, we only use high-quality primary raw materials from first-class manufacturers. We also invest a great deal of time in training our employees in the areas of personal and working hygiene. Last but not least, we have a person who keeps the plant clean from morning to night.

Which areas do you believe offer potential for enabling SAF Bianco to grow further?

Our customers are international brands, and we work well together. We are therefore convinced that we will be able to grow together with our customers. It will not be easy, however. Our competitors are also well equipped, and prices are continually coming under pressure. Generally speaking, we want to grow slowly – firstly, in terms of our products, then, as regards our customers, and, finally, with respect to the areas in which we operate. This will help us avoid errors and ensure we maintain the current high level of quality.

In what respect can SAF Bianco become even better?

We have invested a great deal in recent years and now find ourselves at a very high standard in all areas. Nevertheless, our packaging should still be improved further with a view to the overall product. This is especially true from an environmental perspective. Until now, it has not been possible to use biodegradable products, as neither the transparency nor the





SAF Bianco produces a wide range of packaging for many renowned brands in the domestic food sector.

price of the packaging were of a satisfactory level. We have specific plans in this area, but they still require some further development.

**"WE WANT TO GROW SLOWLY.
THIS WILL HELP US AVOID ERRORS
AND ENSURE WE MAINTAIN THE
CURRENT HIGH LEVEL OF QUALITY."**

Massimiliano Bianco, CEO.

How can Otto Hofstetter AG support you here?

This is still an unknown in this specific case. Generally speaking, however, we expect that our partners follow our requirements. The Italian market is very unique and has its own laws. For us, it is important that we are able to manufacture our products using tools that are extremely precise and very reliable. To enable us to respond quickly to new challenges, we also attach great significance to regular dialogue.

Why are you working with Otto Hofstetter AG?

I am currently reading a book by Simon Sinek entitled

Leaders Eat Last. The book states that a leader opens doors for others and offers them support. One of the strengths of Otto Hofstetter AG is that we both view each other as partners. And partners are friends: a handshake, phone call or e-mail are all it takes to get the ball rolling. We have gathered wide-ranging experience and have seen what others understand by "team". A further aspect relates to social responsibility. Just as we are committed to Italy as a location, Otto Hofstetter AG is committed to Switzerland. Both companies could produce their goods at a lower cost somewhere else in the world, but neither one does so.



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www.safbianco.it

Founded: 1945

Employees: 37

Machinery: 14 injection machines, 3 sleeve systems,
2 offset printing machines



“ONE OF THE STRENGTHS OF OTTO HOFSTETTER AG IS THAT WE BOTH VIEW EACH OTHER AS PARTNERS.”

Massimiliano Bianco, CEO.

What could your partner in Uznach improve further?

For us, it would be great if we were able to meet more frequently. If we were in slightly closer contact and our dialogue was more intensive. At present, we are still a small firm. Thanks to its successful work, Otto Hofstetter AG possesses a great deal of expertise in our area. If we were able to discuss our problems with them more frequently and at an earlier stage, this would be beneficial to both of us. We would be able to grow further, and they in turn could develop more tools for us. Otherwise, we are extremely satisfied with our partner in Uznach. They offer us a great deal of respect and treat us as if we were a big player.

You have officially been the CEO of SAF Bianco since 1 September 2016. What will change under your leadership?

For all intents and purposes, nothing will change. The targeted growth has been achieved together by the second and third generations; all decisions and choices have been made together. In fact, it is only from a legal perspective that there has been a change at the top of the firm. I have actually been responsible for production and the development of the company since back in 1995. The changeover will thus be absolutely

painless. Nothing will change whatsoever for customers, suppliers and other partners. SAF Bianco is both old and young at the same time. Our company is made up of two generations that help each other.

What will the future bring?

Over recent years, design activities have become ever more important, and the issues relating to materials and pricing have remained omnipresent. We will indeed have to find good solutions for tackling these challenges in the near future. In 2017, our turnover will total around EUR 12 million, and we know that we have the potential to generate further growth. However, to do so we will require a larger production area. There are opportunities here and plans are in place. We still do not know what we will be able to put into practice. I cannot look any further into the future. The world – and thus our industry – is changing far too quickly, and the third generation is still much too young to already start thinking about the possible addition of the fourth.

Thank you very much for the interview, Mr Bianco. I wish you continued success in the future.

PET WALK.

Every day, new concepts see the light of day about how to keep after-use PET waste in the materials cycle. International lifestyle brands have embarked on a potential path. They convert waste into shoes.

Various lifestyle and sports brands like Nike, Adidas and Timberland and manufacturers of sustainable textile materials combine several good approaches with cooperation. Here, the emphasis is on ecological aspects, social responsibility and the joy associated with an environmentally aware fashion experience.

One of the suppliers of materials made from recycled PET is Thread International. In Haiti and Honduras, the company employs around 3600 bottle collectors and production workers. Thanks to these efforts, millions of plastic bottles are removed from canals and the roadside and then converted into valuable raw materials. In addition to earning a wage, the employees also benefit from vocational training, improved working conditions and micro-loan programmes.

The shoe manufacturer Timberland even makes it easy for its purchasers to completely trace the history of their new shoes. From the collection of the bottles over recycling and the environmentally friendly production of fibres down to delivery: the history of the Timberland product is illustrated – the customer experiences the impact of his purchase in a completely new light. The PET collection of the outdoor lifestyle brand is scheduled to go on sale in the spring of 2017. The sports article manufacturer Adidas has just put a product on the shelves.



TOLERANCE HALVED.

bischof präzisionsschleifen AG supports Otto Hofstetter AG through the end-to-end manufacture of precision components. The specialist Swiss company guarantees a higher level of precision in series production than scarcely any other supplier. The shared passion for the highest quality has turned the two companies into a successful team.

Diepoldsau, a village with just over 6000 inhabitants, is sandwiched between the Rhine and the Swiss-Austrian border. The region ranks amongst the most important industrial conurbations in Switzerland and is the location of numerous internationally renowned companies. At first sight, the building of bischof präzisionsschleifen AG does not really stand out from the neighbouring companies. However, once you cross the threshold, it feels like you have stepped into the hangar of an aircraft manufacturer or the workshop of a Formula 1 racing team: light-filled rooms, shiny floors and an extremely pleasant climate. "Two years ago we increased in size, modernised and invested in energy efficiency", explains owner and CEO Alexander Bischof.

Presentation of the company.

He set up the company in 1994 starting with a grinding machine in the basement of a private house. The company focused on flat and profile grinding and attracted more and more customers thanks to its specialisation. One of its very first customers was Otto Hofstetter AG which had articles manufactured by Alexander Bischof. After an initial expansion phase in 1999, the success chalked up two years ago paved the way for further growth. The already progressive infrastructure was further modernised and measures implemented for energy efficiency, climate control and solar power. Today, a total of 10 experts work for the company. Production is in full swing with 10 grinding machines and 5 milling machines. With a five-axis milling machine with integrated loading robot, the forward-thinking entrepreneur Alexander Bischof has already

taken the first steps towards Industry 4.0. "If you want to continue to be successful, you have to constantly optimise and strike out in new directions."

Technical speciality.

In other areas, too, Alexander Bischof constantly has his eye on the ball. Besides extremely precise CNC machines and qualified, motivated employees, he only uses high-quality grinding discs and optimises existing processes. If a device in the machines fails to meet his requirements, the deficit is promptly remedied with an in-house design solution. This constant striving for the best possible solution has also turned him into a sought-after specialist for creep-feed grinding. Hardly any other supplier uses this method. His technique is unique particularly when it comes to creep-feed grinding a groove. When a manufacturer uses the pendulum grinding technique, the groove or profile is conical. This means that the interlinking parts have more slack. This encourages wear and reduces the quality of the product. With creep-feed grinding this can be considerably improved. For example, a 20-millimetre groove is ground with a feed rate of 30 to 50 millimetres per minute. For a component with a length of 200 millimetres this means an operating time of five minutes. Manufacturing an identical groove using a pendulum grinding technique takes around two hours. The creep-feed grinding technique is, therefore, much faster and, at the same time, far more precise.

Precision and economic viability.

The more perfect the component, the higher the price – that

A close-up portrait of Alexander Bischof, a man with short, dark hair and a light beard, smiling and looking slightly to the side. He is wearing a dark blue button-down shirt. The background is a soft, out-of-focus grey.

"THE PARTNERSHIP WITH
OTTO HOFSTETTER AG
IS A WONDERFUL EXAMPLE OF
SUCCESSFUL COOPERATION."

Alexander Bischof, CEO.

is the normal rule. "With us, the opposite is the case", explains Alexander Bischof. Thanks to the precision his company strives for, they can produce more economically. Where normally two to four parts are worked simultaneously, in the case of Bischof there are series of six to eight. To meet or even exceed customer specifications, the specialists halve the stipulated tolerance internally. For instance, for the slide insert in a mould manufactured by Otto Hofstetter AG. Eighteen operations are needed for the complete manufacture of this component. The desired tolerance is zero minus one-hundredth millimetre. "We work internally with half the tolerance specified by the technicians from Uznach, in other words with just five μm ." This halved standard deviation made it possible for us to manufacture several parts at the same time. Production is faster, more economical and perfectly within the set tolerance values.

Optical quality.

Besides the technical quality, optical quality is important for Bischof, too. When the company manufactures components,

it always focuses on external appearance as well. "A part that leaves our production site always looks optically perfect." Alexander Bischof admits that appearance to a large degree only contributes subjectively to quality. "Nonetheless, the esteem for our work is higher when all the edges are smooth and all areas are absolutely scratch-free." The downstream companies have heard about this attitude. Components from the workshop in Diepoldsau are treated with corresponding care.

"PRECISION IS AN ATTITUDE AND A MINDSET."

Alexander Bischof, CEO.

Understanding of cooperation.

"Added value should result in a three-fold manner from this cooperation", says Alexander Bischof with conviction. The product must gain in quality and the customer must be faster, more flexible and more precise. "To comply with these two factors, it is important that we have an input and therein lies the added value for us." If bischof präzisionsschleiferei AG is

"THE EXCHANGE IN ALL AERAS IS ALMOST AS SIMPLE AS IF WE WERE NEXT DOOR."

Alexander Bischof, CEO.



For Alexander Bischof, mutual trust and the exchange of knowledge are the mainstays of a solid partnership.

given an opportunity to offer advice to its customer, it will become part of the process and its quality. This turns each opportunity for cooperation into a win-win situation.

Cooperation with OHAG in concrete terms.

Although the mighty Appenzell Alps lie between Diepoldsau and Uznach, cooperation is so smooth you would think they were neighbours. Upstream and downstream partners like the companies specialising in the hardening of parts or the coating of components are located mid-way to Uznach. That is the logistics side. "At Otto Hofstetter AG, perfect quality has the highest priority. As it does for us." This already puts the partnership on a sound footing. Alexander Bischof is convinced of this. The challenges facing a mould today are enormous. This starts with the choice of material and continues with the way in which a component is manufactured. When the design engineer sketches a component, there are many different options in terms of precision and the related tolerances. According to Alexander Bischof, defining them always has something to do with the efforts involved in production. "For that reason we are already invited on board by our colleagues in Uznach in this early stage of the project."

From partners to friends.

Over a period of 22 years you jointly experience many highs as well as lows. This creates a bond. For companies, too. Business meetings are supplemented by private get-togethers. "Our business does not suffer from this. The opposite is the case", stresses Alexander Bischof. Openness and solidarity ensured in January 2015 – when the Swiss National Bank lowered the Euro exchange rate – that both companies rapidly came up with a response to the currency-driven inflation of the prices for moulds of Otto Hofstetter AG. "Today, when society is dominated by a frantic pace and a non-committal attitude, the constructive and friendly discussions with the management of Otto Hofstetter AG are a gift from heaven."



Ongoing investment in staff and infrastructure ensure bischof präzisionsschleifen AG is in a position to offer the precision required of it by Otto Hofstetter AG.



bischof

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Partner for system solutions incorporating end-to-end manufacture of precision components for companies in the fields of mould manufacturing, automated plant construction and the automotive industry.

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